The Bronx High School of Science
M. Rachel Hoyle, Principal

Mathematics Department
Kathleen Lyons, Assistant Principal

Name: Rafan Rayan

Date: 5-7-2026

Geometry Homework 92 – Coordinate Geometry

Aim: Be able to scale line segments.

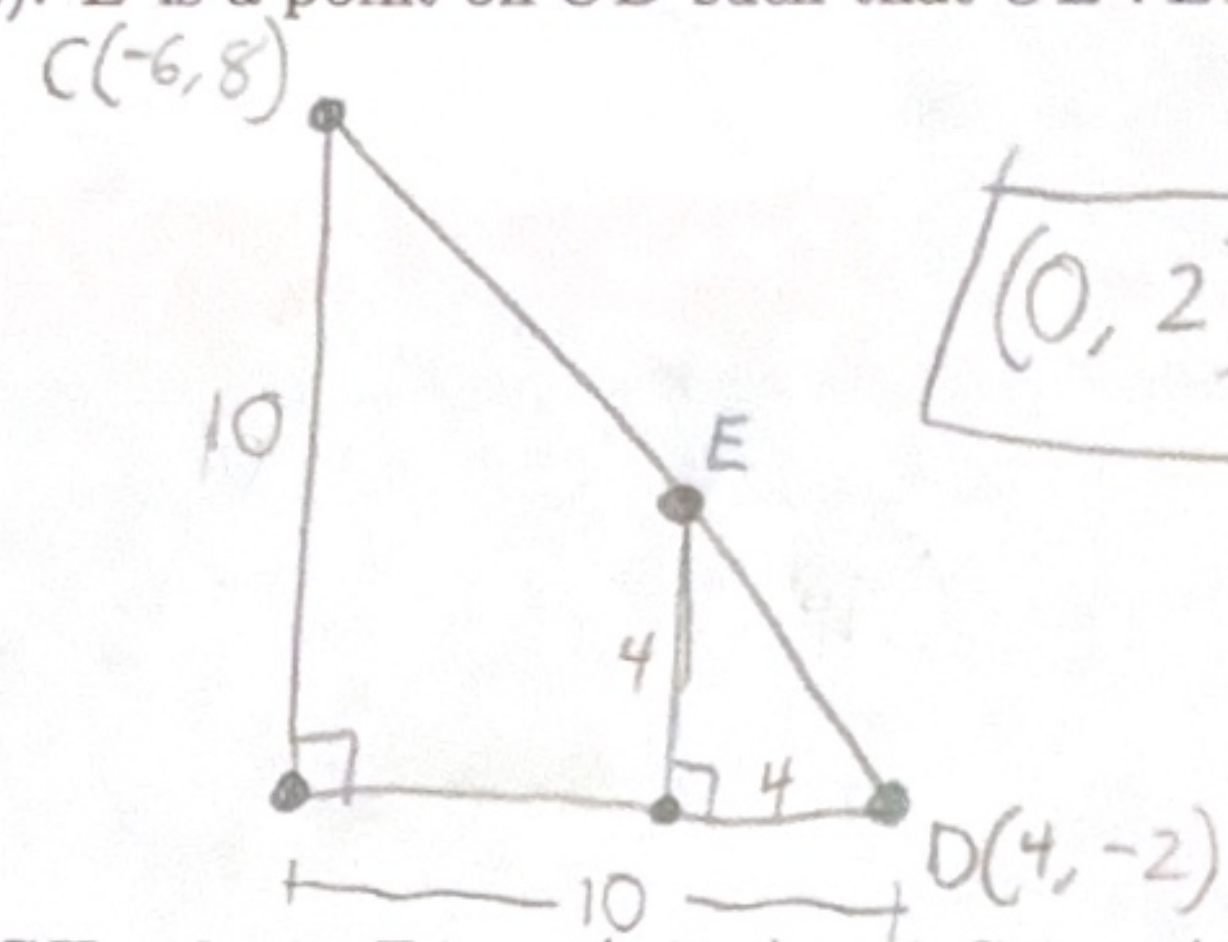
1. $\overline{AB}$ is drawn such that A is at $(3x+5, 3y)$ and B is at $(x-1, -y)$. Find the coordinates of M, the midpoint of $\overline{AB}$, in terms of x and y.

$$\frac{3x+5+x-1}{2} = \frac{4x+4}{2} = 2x+2$$

$$\frac{3y-y}{2} = \frac{2y}{2} = y$$

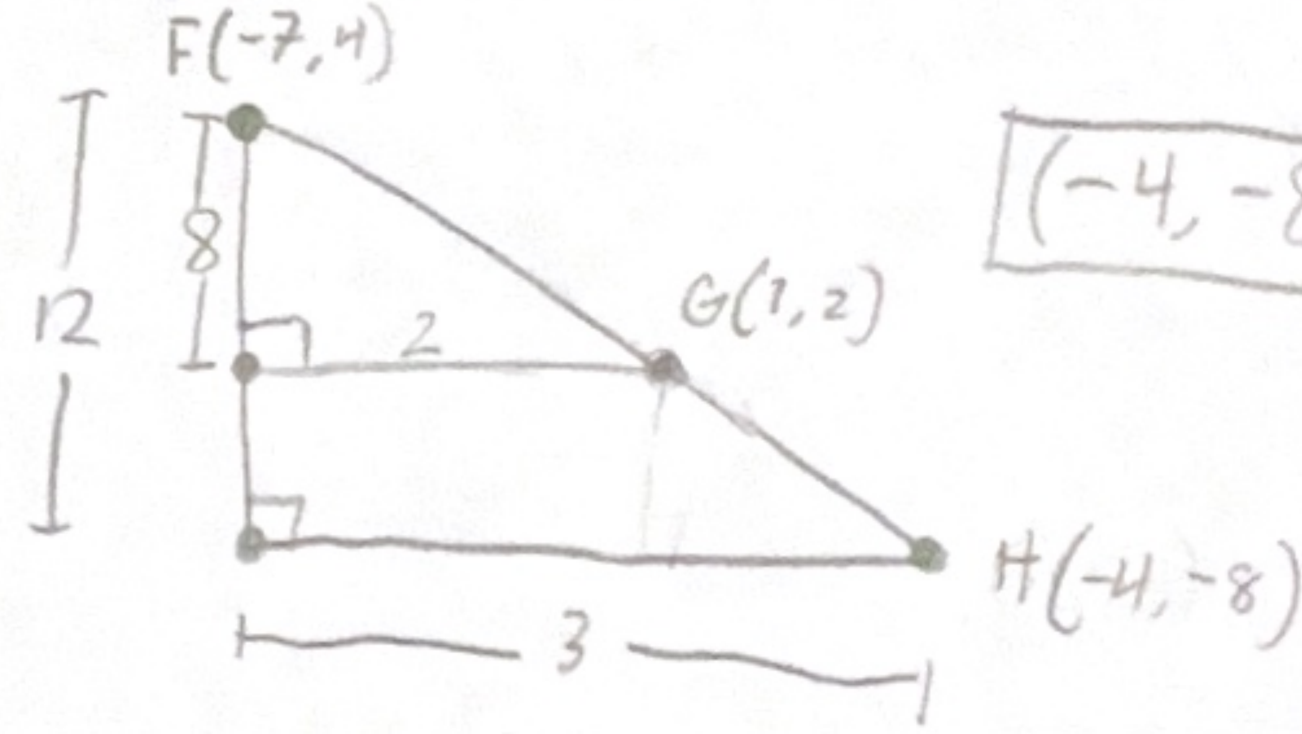
$$\boxed{(2x+2, y)}$$

2. $\overline{CD}$ is drawn such that C is at $(-6, 8)$ and D is at $(4, -2)$. E is a point on $\overline{CD}$ such that $CE : ED = 3 : 2$. Find the coordinates of E.



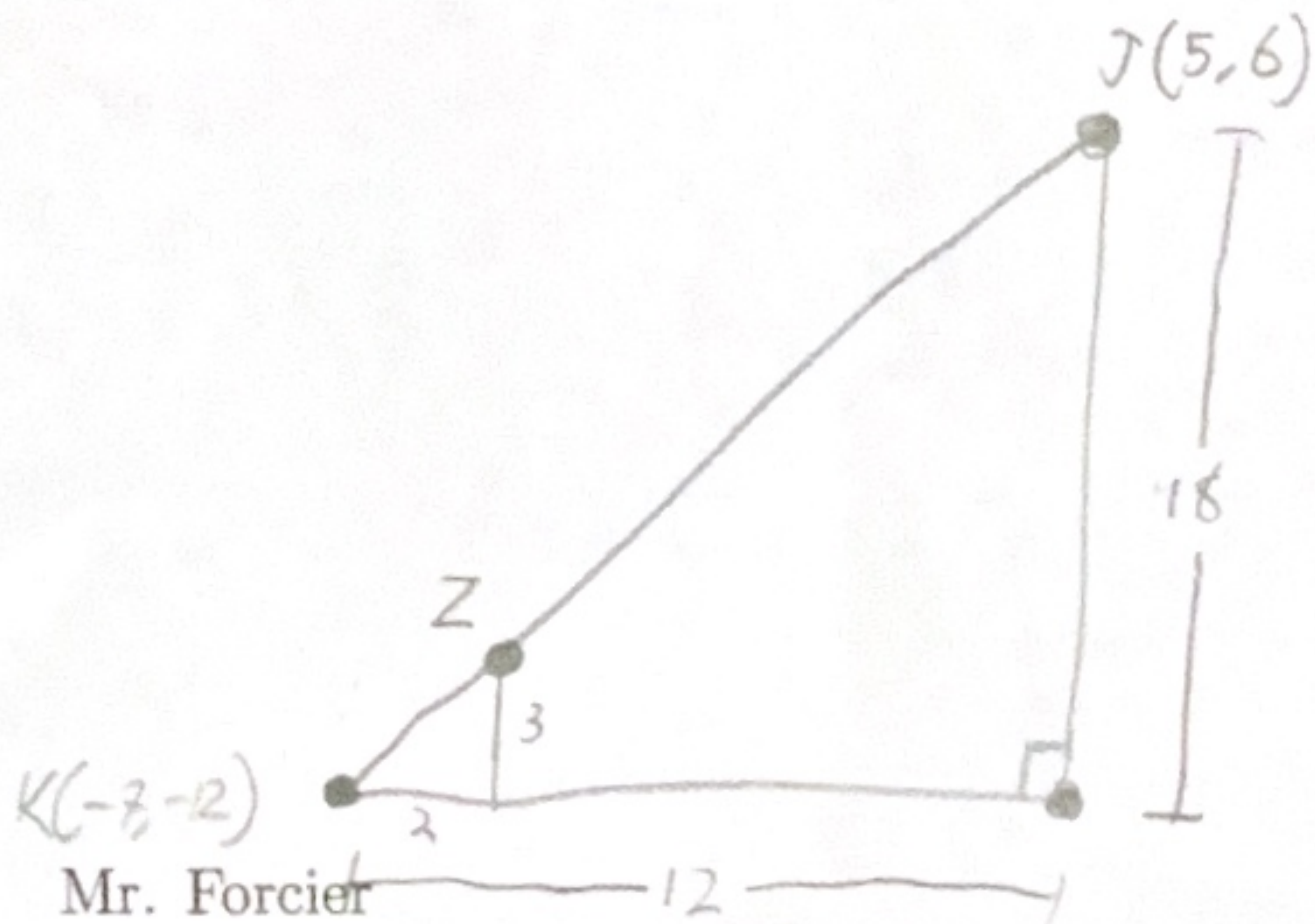
$$\boxed{(0, 2)}$$

3. $\overline{FG}$ is extended through G to a point H such that $FG : GH = 2 : 1$. F is at $(-7, 4)$ and G is at $(1, 2)$. Find the coordinates of H.



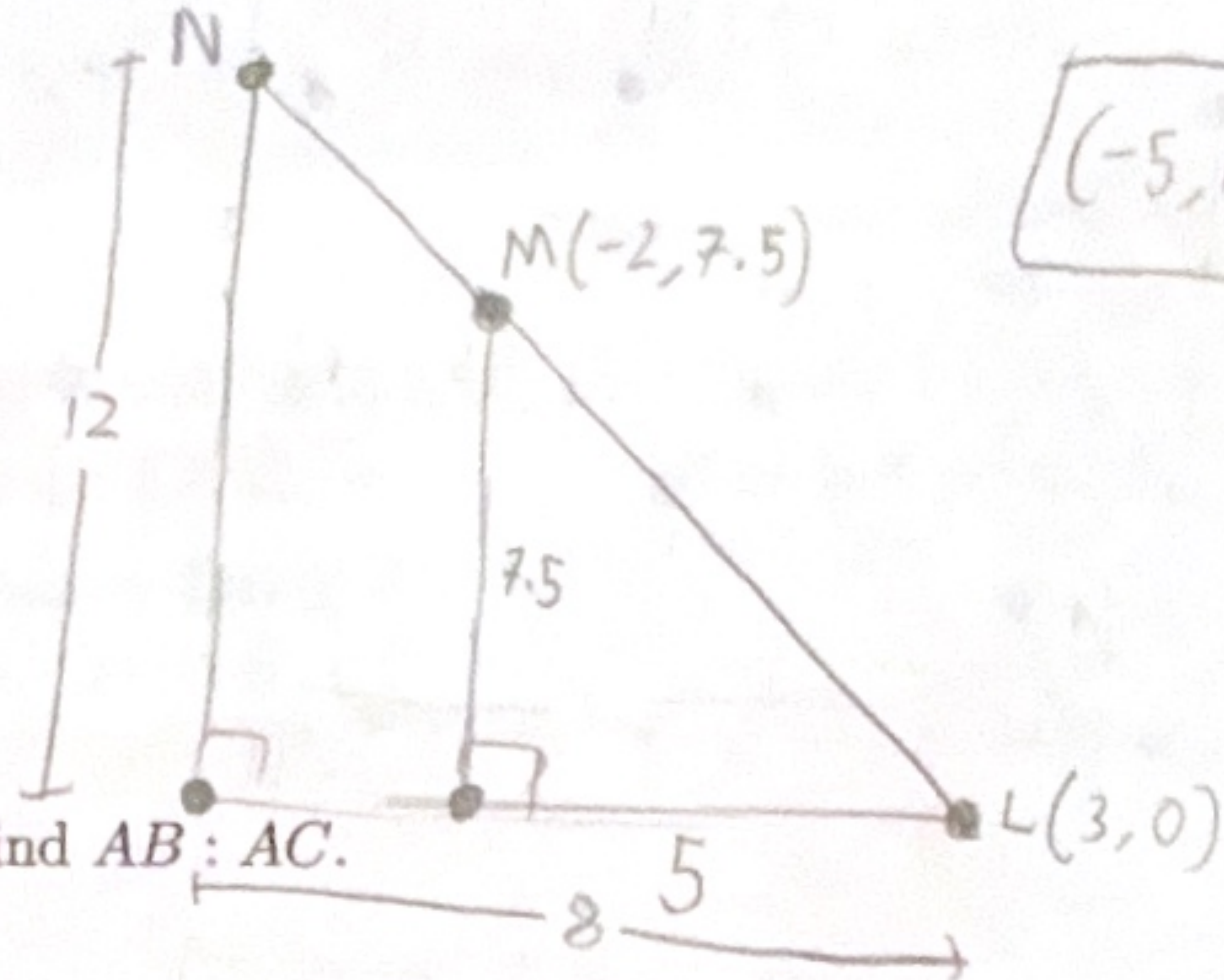
$$\boxed{(-4, -8)}$$

4. $\overline{JK}$ is drawn such that J is at $(5, 6)$ and K is at $(-7, -12)$. Z is a point on $\overline{JK}$ such that $JZ : ZK = 5 : 1$. Find the coordinates of Z.



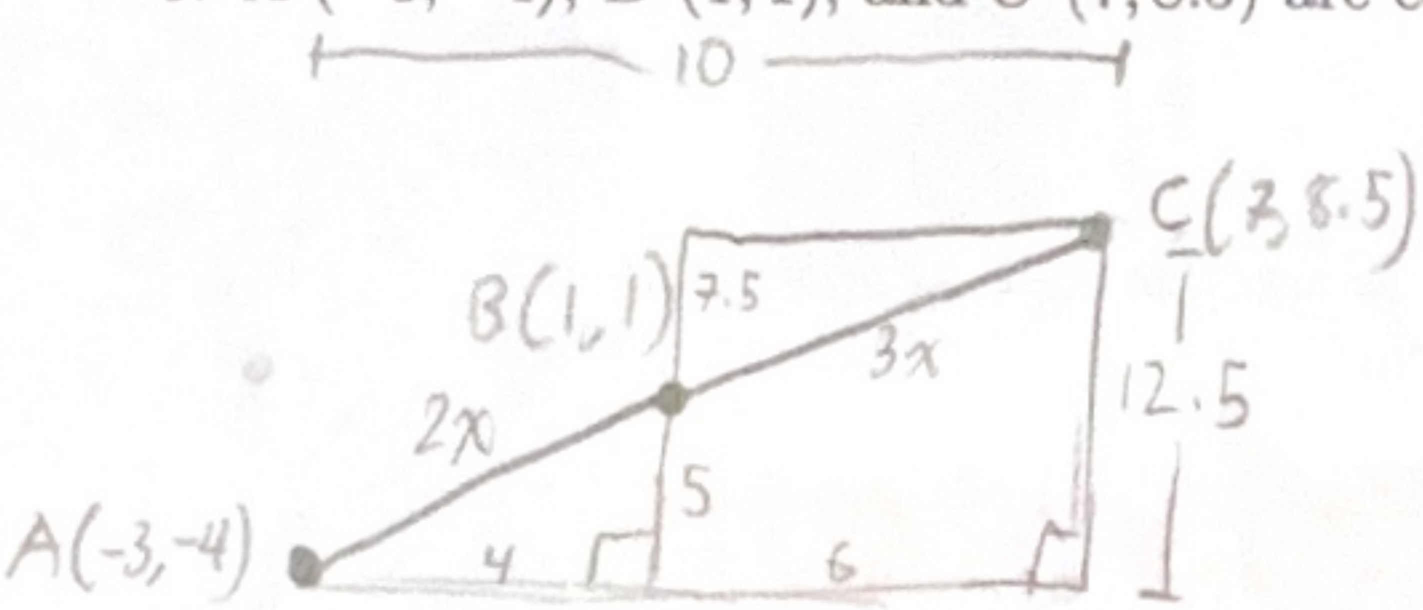
$$\boxed{(-5, -9)}$$

5. $\overline{LM}$ is extended through M to a point N such that $LM : MN = 5 : 3$. L is at $(3, 0)$ and M is at $(-2, 7.5)$. Find the coordinates of N.



$\boxed{(-5,12)}$

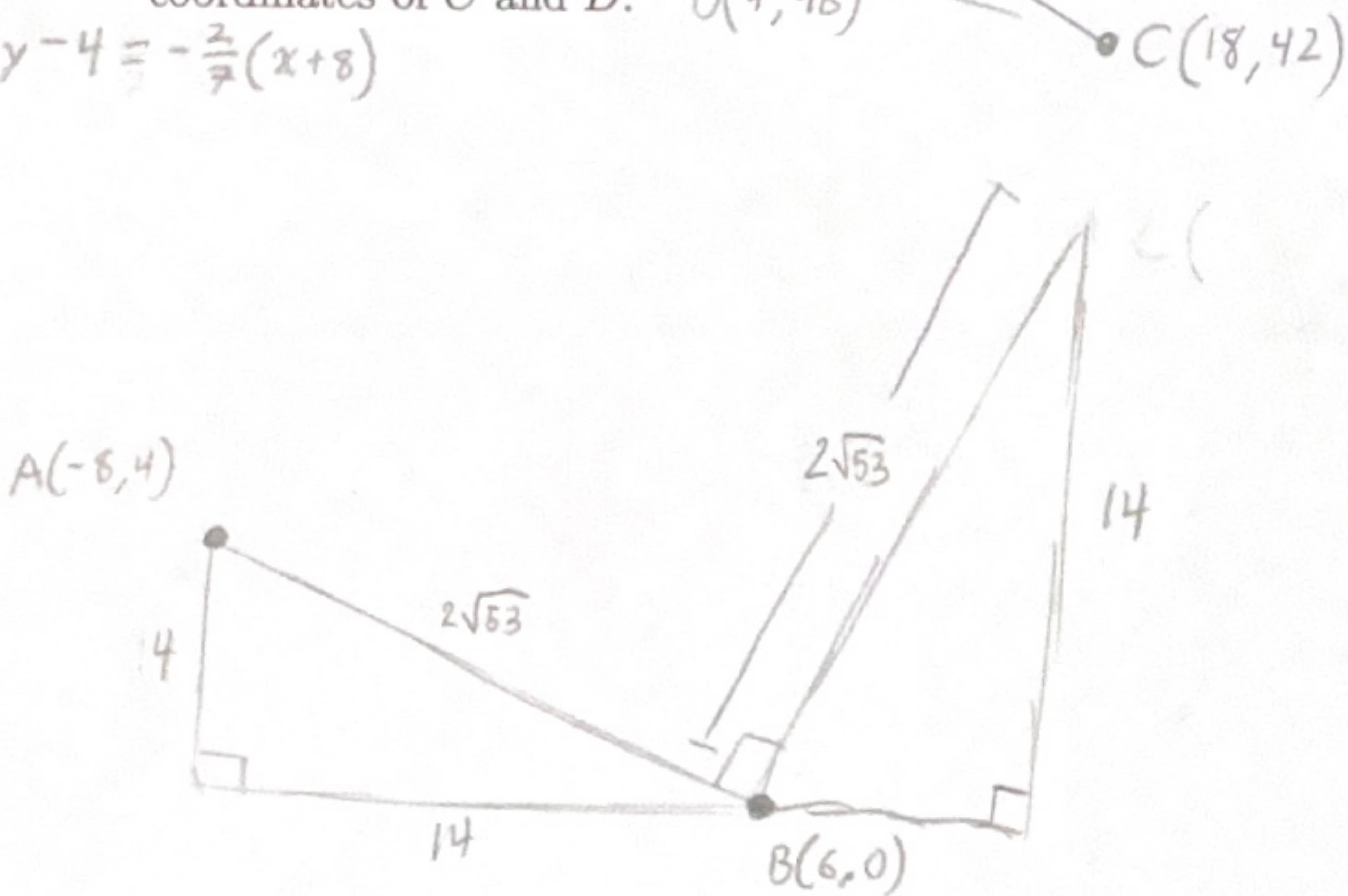
6. A $(-3, -4)$, B $(1, 1)$, and C $(7, 8.5)$ are collinear. Find $AB : AC$.



$\boxed{2:5}$

7. Rectangle $ABCD$ is drawn such that A is at $(-8, 4)$, B is at $(6, 0)$, and $AB : BC = 1 : 3$. Find the coordinates of C and D.

$AB: y-4=-\frac{2}{7}(x+8)$



$\boxed{\begin{array}{l} C=(18,42) \\ D=(4,46) \end{array}}$

8. Write equations of the lines that contain $\overline{AB}$ and $\overline{BC}$ in problem 7.

$\overline{AB} \longrightarrow y-4=-\frac{2}{7}(x+8)$

$\overline{BC} \longrightarrow y-42=\frac{7}{2}(x-18)$